

“THE IDEA OF AGENTS IS NOT NEW; WE’VE BEEN WORKING ON THIS FOR A WHILE... BUT THE REASON WHY IT’S GETTING SO MUCH ATTENTION NOW IS BECAUSE LARGE LANGUAGE MODELS AND GENERATIVE AI ACCELERATED SOME OF THE CHARACTERISTICS AGENTIC AI NEEDS TO BE SUCCESSFUL.”



— **THADEOUS GOODWYN, DIRECTOR OF GENERATIVE AI,**
BOOZ ALLEN HAMILTON (IN A MEDIA INTERVIEW)

information on prescribed medication usage, appointment scheduling, and more.

- **Content creation.** Agentic AI can help create high-quality, personalised marketing content, freeing up the marketers’ time to focus on strategy and innovation.

- **Data analysis.** AI agents are very good at analysing video, audio, images, text, or any other form of data to extract useful information—whether it is skimming through security footage to extract details of a crime, or monitoring high-altitude climbers using critical health parameters relayed from wearables and video feeds from drones, or monitoring financial transactions to detect and avert threats.

- **Software development.** Agentic AI can help manage the entire software development lifecycle, from system design to writing and debugging code and quality assurance. More commonly, software companies are using AI agents for repetitive coding tasks, enabling developers to focus on more complex challenges. In recent interviews, experts like Bill Gates, co-founder of Microsoft, Sam Altman, CEO of OpenAI, and Sridhar Vembu, founder and former CEO of Zoho Corporation, have pointed out that skilled software professionals will still be in demand in the foreseeable future to handle complex coding challenges, which cannot be typecast.

“When people say ‘AI will write 90% of the code,’ I readily agree

because 90% of what programmers write is ‘boilerplate.’ There is ‘essential complexity’ in programming, and then there is a lot of ‘accidental complexity’ (that is the boilerplate stuff), and this is very old wisdom from the Mythical Man Month. AI is doing a great job of eliminating the accidental complexity. Humans are still needed to deal with the essential complexity. In essence, AI can make mincemeat of patterns already discovered (by humans). Can it find totally new patterns? As with humans, that is much rarer and a quality known as ‘taste’ or ‘knowing where to dig’ or ‘follow a hunch or conviction all the way’ is needed to discover new patterns. I don’t know if AI can do this. I don’t know if that can be brute-forced,” wrote Vembu on X.

- **Autonomous vehicles.** Agentic AI helps autonomous vehicles perceive and understand the environment, navigate roads, make decisions in real time, and ensure safe driving.

- **Financial services.** Agentic AI is being used in the banking, financial services, and insurance (BFSI) industry for fraud detection, risk assessment, and customer onboarding. It can automate routine tasks like expense reporting, compliance checks, financial reporting, and invoice processing. It can monitor market trends, assess investments and credit risks, and advise customers based on their financial history and goals.

- **Research and development.** AI agents can be helpful research assistants, as they can analyse existing knowledge, explore possibilities based on user-defined characteristics, and suggest new possibilities. They are especially helpful in drug discovery and material sciences.

- **Human resources management (HRM).** The use of agentic AI in HRM is picking up all around the world. It is used for tasks like initial application screening, interview scheduling, onboarding, payroll processing, personalising benefits recommendations, evaluating leave requests, managing document workflows, assisting employees with information about policies and benefits, and more.

- **Defence and military logistics.** Agentic AI helps plan the movement of defence personnel and equipment via multiple transport modes. In the defence sector, this is a very critical task, as it needs to be done quickly and efficiently during emergencies, when every minute makes a difference.

- **Manufacturing.** AI agents can be super helpful in long manufacturing workflows, spanning from procurement to production. While the simplest of AI agents can detect low material stock and alert the concerned stakeholders, a full-fledged agentic AI system can proactively detect the contingency, and pre-empt it by finding alternate suppliers, placing orders, and reconfiguring production schedules!

- **Utilities.** Agentic AI is used in the utilities industry for various tasks like infrastructure repairs and maintenance, constant monitoring and management to prevent outages, assessing and organising responses to disasters, ensuring regulatory compliance, and most importantly, customer service. In some countries, for example, there is a regulatory requirement for utility